Abdominal Pain

Abdominal Examination
- Inspect for distension, bruising, or visible abnormalities.
- Palpate for tenderness, rigidity, or masses.
- Note rebound tenderness or guarding.

Possible Causes of Abdominal Pain
- Myocardial infarction Pneumonia/PE
- Pregnancy (including ectopic)
- CHF
- Aortic dissection
- Aortic aneurysm
- Peptic ulcer disease
- Kidney stone
- Pancreatitis/Gallbladder disease Bowel obstruction
- GI Bleeding/Appendicitis

Consider Red Flags
- Severe, unrelenting pain
- Fever (>38°C)
- Persistent vomiting or hematemesis
- Suspected pregnancy complications (e.g., ectopic pregnancy, miscarriage)

Assess for 12-Lead ECG Requirement

Perform 12-Lead ECG for patients who:
- Are greater than 40 years of age, and/
  - Has 3 or more cardiac risk factors, including:
  - Smoking (current or cessation within last 3 months)
  - Obesity
  - Diabetes
  - Hypertension
  - High cholesterol
  - Cardiovascular disease
  - Family history of cardiovascular disease

If Signs of Shock or Dehydration

Fluid Resuscitation
- Administer Normal Saline or Ringers Lactate:
  - Adults: 500-1000 mL bolus, reassess, repeat as needed
  - Pediatrics: 20 mL/kg bolus, reassess, repeat as needed

EMT

Assess & Manage Pain

Go to Pain Management CPG

Abdominal Cramping Or Menstrual Cramps

Consider Hyoscine Butylbromide
Adults Only: 20mg IV/IM

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If Nausea & Vomiting Present

Go to Nausea & Vomiting CPG

Patient Care Goals

1. Identify life-threatening causes of abdominal pain
2. Improve patient comfort

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Abdominal pain or discomfort, related to a non-traumatic cause

Exclusion Criteria

- Abdominal pain due to trauma
- Abdominal pain related to pregnancy

Key Patient Assessment Points

1. **Abdominal Examination**
   - Inspect for distension, bruising, or visible abnormalities.
   - Palpate for tenderness, rigidity, or masses.
   - Note rebound tenderness or guarding.
2. **Patient history**
   - Use OPQRST and SAMPLE history to identity likely cause.
   - Investigate patient history to exclude possibility for pregnancy.
3. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary and Secondary Survey**
   - Perform primary and secondary survey as per Universal Care CPG.
2. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
3. **Consider Red Flags**
   - Severe, unrelenting pain
   - Fever (>38°C)
   - Persistent vomiting or hematemesis
   - Suspected pregnancy complications (e.g., ectopic pregnancy, miscarriage)
4. **Evaluate for Signs of Shock or Instability**
   - Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics)
   - Tachycardia (HR >100 in adults, or age-specific for pediatrics)
   - Altered mental status
   - Poor perfusion (cool, clammy skin, delayed capillary refill)
5. **Assess for 12-Lead ECG Requirement**
   - Perform 12-Lead ECG for patients who:
     - Are greater than 40 years of age, and/
     - Has 3 or more cardiac risk factors, including:
       - Smoking (current or cessation within last 3 months)
       - Obesity
       - Diabetes
       - Hypertension
       - High cholesterol
       - Cardiovascular disease
       - Family history of cardiovascular disease
6. **Establish IV Access**
   - Indicated for dehydration, hypotension, or anticipated medication administration
7. **Fluid Resuscitation**

- Indications:
  - Signs of Shock or dehydration.
- Administer Normal Saline or Ringers Lactate:
  - Adults: 500-1000 mL bolus, reassess, repeat as needed
  - Pediatrics: 20 mL/kg bolus, reassess, repeat as needed

8. **Assess and Manage Pain**
   - Assess and manage pain as per Pain Management CPG.
9. **If Nausea/Vomiting Present**
   - Assess and manage as per Nausea & Vomiting CPG.
10. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Transport/Transfer Considerations:
  - Abdominal pain can constitute a medical emergency which is poorly detected in the ambulance setting.
  - All patients with non-specific abdominal pain or persistent vomiting should be recommended to see a doctor as soon as possible (if transport to hospital is refused).
  - Patients with early signs of shock, unstable vitals, or a serious cause of abdominal pain is suspected (e.g. appendicitis) or a serious cause cannot be ruled out should be routinely transported to hospital.

Clinical Insights & Additional Guidance

- Common causes: gastroenteritis, appendicitis, bowel obstruction, pancreatitis, kidney stones, ectopic pregnancy, DKA, or mesenteric ischemia.
- Possible Alternative causes: Myocardial infarction, Pneumonia/Pulmonary Embolus, Pregnancy (including ectopic), CHF, Aortic dissection, Aortic aneurysm, Peptic ulcer disease, Kidney stones, Pancreatitis /Gallbladder disease, Bowel obstruction, GI Bleeding, Appendicitis.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Abdominal pain, nausea, vomiting – include duration, severity, and any relieving or aggravating factors.
- Injuries / Information: Relevant history – recent food intake, medications, bowel habits, associated symptoms (e.g., fever, diarrhea, hematemesis).
- Signs: Vital signs, abdominal exam findings (tenderness, guarding, rebound), mental status.
- Treatment: Medications given (e.g., metoclopramide, paracetamol), IV fluids, pain scale before and after intervention.
- Allergies: Particularly medication-related (e.g., antiemetics or analgesics).
- Medications: Current prescribed or OTC medications.
- Background: History of GI disease, renal issues, diabetes, prior surgeries.
- Other: ECG findings if performed, pregnancy status if applicable, transport rationale.

Key Performance Indicators

1. Assess and document initial and post-treatment pain score.

References

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9. Schwerin, D.L. and Mohr, N.M. (2023) 'EMS Pain Assessment And Management', in *StatPearls* [Internet]. Treasure Island (FL): StatPearls Publishing.